

FOOD FOR THOUGHT



Extreme weather events, rising temperatures, and floods are impacting food production



Food inflation has become persistent and has pushed up headline retail inflation



Its impact is evident on prices of vegetables, cereals, pulses and oilseeds

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IF YOU HAVE been keeping your air conditioners and fans on for longer and are paying more for food items in recent years, it's interconnected. This is because rising temperatures have not only made the summer months longer and hotter but have also impacted food production—increasing

everything from your electricity to grocery bills—a stark reminder that climate change is upon us.

Scientists have been warning for long that the earth's surface temperature is rising, but it appears to be increasing at a faster pace now. Consider the evidence: according to official data, average mean surface temperature in India has risen from 25.61° C in 2012 to 26.15° C in 2023. The mercury in the last few summers has touched and even crossed the 50° C mark in several Indian cities. The annual average global temperature in 2023 was about 1.5° C above the pre-industrial levels (1850–1900). According to the India Meteorological Department's (IMD) annual report 2023, the annual mean land surface air temperature averaged over India during 2023 was +0.65° C above the long-term average (1981–2010 pe-

riod). "The year 2023 was the second warmest year on record since nationwide records commenced in 1901," it noted.

Scientists and economists agree that rising surface temperatures have impacted yields—from vegetables and durable crops to even animal sources of protein—leading to a sustained increase in food inflation. And if this isn't enough, unseasonal rains, seen as another sign of climate change, have added to the woes, further fuelling inflationary pressures.

"Food production is dependent on rains and temperature. The weather patterns have changed in the past few years, and there is more volatility in both temperature and rainfall," notes D.K. Joshi, Chief Economist at rating agency CRISIL.

The year 2023 was the hottest

TROUBLE AHEAD?

A study by the Indian Council of Agricultural Research shows the likely impact of climate change on food yields without adaptation and mitigation measures:

20%

fall in rain-fed rice yields in 2050 and 47% in 2080 scenarios

3.5%

reduction in irrigated rice yields in 2050 and 5% in 2080 scenarios